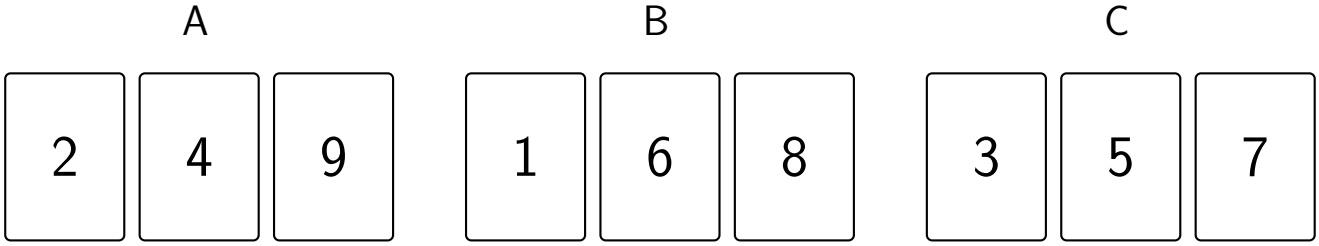
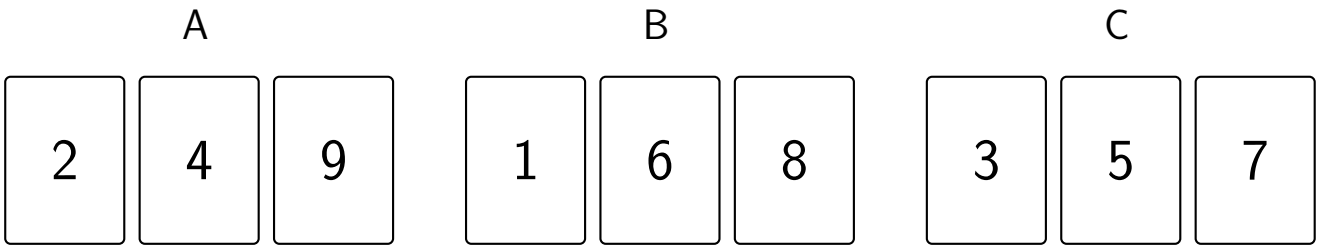
Each physical card is equally likely. Mix each deck separately and draw one card from each. Return both cards before the next round. The larger number wins. The two decks in a comparison must not share a number. Card order in a deck does not matter.

Problem 1: For each pair of decks, find which deck wins more of the possible card pairs. Is there a deck that wins more pairs than both of the others?



Problem 2: Two bags hold two different decks from A, B, and C. You hear only which bag wins. The left bag wins six draws in a row. Can you be sure it wins more possible card pairs? Could any fixed number of these results make you sure? Explain.



Problem 3: Keep 9 in A, 8 in B, and 7 in C. Place cards 1 to 6 so A wins more pairs than B, B more than C, and C more than A. Find two arrangements different from Problem 1.

